Process Characterization Master Plan

A-Mab Drug Substance — PCMP-001

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Role	Function	Name (synthetic)	Signature	Date
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Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2025-08-04	Initial issue
2.0	2026-09-30	Re-analysis under INV-001: a-priori intermediate acceptance criteria, fragments by nrCE-SDS, two-sided capability for acidic variants, scale-down models qualified by their offsets, execution dates recorded (see INV-001)

Abbreviations

Abbreviation	Meaning
AEX	anion exchange chromatography
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CEX	cation exchange chromatography
Cpk	process capability index
CPP	critical process parameter
CQA	critical quality attribute
DS	drug substance
GMP	good manufacturing practice
GPP	general process parameter
HCP	host cell protein
HMW	high molecular weight species
IAC	intermediate acceptance criterion
KPP	key process parameter
LMW	low molecular weight species
LRF	log reduction factor
MSAT	Manufacturing Science and Technology
MVM	minute virus of mice
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
PPQ	process performance qualification
QA	Quality Assurance
RSM	response surface methodology
SEC	size exclusion chromatography
SOP	standard operating procedure
UF/DF	ultrafiltration and diafiltration

Abbreviation	Meaning
WC-CPP	well-controlled critical process parameter
XMuLV	xenotropic murine leukemia virus

1 Purpose and scope

This plan governs the Stage 1 process characterization of the A-Mab drug substance process. It sits under the Process Transfer Plan PTP-001, which moves the process from the development site in Cambridge, MA, to the commercial drug substance facility in Grafton, WI. The characterization studies provide the evidence for the operating ranges and the control strategy that are transferred.

The plan covers the 8 steps of the drug substance train listed in Table 1.1, from the production bioreactor (Step 3) to ultrafiltration / diafiltration (Step 10). The seed expansion steps and the drug product process are outside its scope.

Table 1.1: The drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification, where fragments form during the harvest hold
5	Protein A Chromatography	Capture; principal HCP and DNA clearance; source of leached Protein A
6	Low-pH Viral Inactivation	Low-pH hold; inactivates enveloped virus (XMuLV credit)
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; clears HCP, DNA and virus (XMuLV and MVM credit)
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

This plan sets out the critical quality attribute (CQA) framework (Section 3), the risk basis (Section 4), the common approach to scale-down models (Section 5) and to statistics (Section 6), and the framework of acceptance criteria (Section 7) for the whole train. Each Process Characterization Plan (PCP-003 to PCP-010) applies this framework to one step. It adds what is specific to that step, such as the scale-down model, the factors and their ranges, and the sampling plan. Where a step plan departs from this plan, it should state the departure and the reason for it.

The documents of the package build on each other in sequence. The risk assessment RA-001 sets the scope of each study. The Process Characterization Plan of the step

(PCP-00N) plans the study, and the paired Process Characterization Report (PCR-00N) reports it. The Process Characterization Master Report PCMR-001 then consolidates the reports into the control strategy for the whole process. The QA investigation INV-001 reviewed PCMR-001 Version 1.0, and its findings led back to this revision and to the reissue of the other documents. Table 1.2 lists the package.

Table 1.2: Documents of the A-Mab characterization package.

Subject	Document ID	Document class
A-Mab Drug Substance	PTP-001 RA-001	Process Transfer Plan Pre-Characterization Process Risk Assessment
Production Bioreactor (Step 3)	PCP-003 PCR-003	Process Characterization Plan Process Characterization Report
Harvest and Clarification (Step 4)	PCP-004 PCR-004	Process Characterization Plan Process Characterization Report
Protein A Chromatography (Step 5)	PCP-005 PCR-005	Process Characterization Plan Process Characterization Report
Low-pH Viral Inactivation (Step 6)	PCP-006 PCR-006	Process Characterization Plan Process Characterization Report
Cation Exchange Chromatography (Step 7)	PCP-007 PCR-007	Process Characterization Plan Process Characterization Report
Anion Exchange Chromatography (Step 8)	PCP-008 PCR-008	Process Characterization Plan Process Characterization Report
Small-Virus Retentive Filtration (Step 9)	PCP-009 PCR-009	Process Characterization Plan Process Characterization Report
Ultrafiltration / Diafiltration (Step 10)	PCP-010 PCR-010	Process Characterization Plan Process Characterization Report
A-Mab Drug Substance	PCMR-001	Process Characterization Master Report
Acceptance criteria for proven-acceptable-range analysis set after data review	INV-001	Investigation Report

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2025-08-04, before the first characterization study started on 2025-08-27. Version 2.0 replaces it and is issued under INV-001. QA opened the investigation on 2026-08-03 after its review of PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. INV-001 found that the criteria used to judge each step in the proven acceptable range (PAR) analysis were not in any approved protocol and had been set after the data were seen (INV-001 finding 1). Of the 15 Version 1.0 step criteria, the 6 aggregate and host cell protein criteria had been carried back from the drug substance limit through the nominal clearance of this campaign and then divided by a margin that differed by step (INV-001 finding 2). Version 1.0 of this plan defined no criteria for step outputs.

This revision makes the following changes.

1. It defines intermediate acceptance criteria (IACs) for the step outputs on which a drug substance criterion depends (Section 7.3). The rule that derives them uses no data from the studies it judges. They replace the Version 1.0 step criteria described above (INV-001 findings 1 and 2). They also replace the bioreactor limits for the glycans and acidic variants, which were computed from the capability data they were then used to judge (INV-001 finding 3).
2. It defines how each report verifies the platform worst cases that the criteria rely on, and when that verification may change a criterion (Section 7.5). Verification tightens a criterion where a step does worse within its NOR than the platform worst case. It widens a criterion only where the 95 % confidence bound of the A-Mab worst case is better than the platform value and the model behind it is adequate. In both directions the bound, not the point prediction, replaces the platform value.
3. It qualifies each scale-down model by its offset from manufacturing scale, not by a test of equivalence, and states every judgement against a criterion at manufacturing scale (Section 7.1).
4. It sets non-reduced CE-SDS (nrCE-SDS) as the method for fragments at every step, following the decision of 2026-03-20 to change the fragment method to nrCE-SDS (AMD-001, Section 7.8). This answers INV-001 finding 6, that size exclusion chromatography (SEC) reads fragments high in samples that still carry host cell protein.
5. It assesses capability for acidic variants against both ends of the criterion (Section 6.5). Version 1.0 assessed it against the upper end only (INV-001 finding 4).
6. It judges viral clearance at the worst case within the NOR (Section 6.4). That is the mean predicted log reduction with the other parameters at their least favourable combination of NOR low, set-point and NOR high. Viral clearance is validated at worst-case conditions, so the claim must hold there. Quality attributes are judged on a single batch. Version 1.0 judged every response on a single batch.
7. It records the execution window of every study and the calibration status of the equipment each study used (Section 5 and Section 9). The Version 1.0 records

gave no execution dates, so the equipment tables appeared to cite expired calibrations (INV-001 finding 5). The Version 1.0 title blocks of the plans also carried 2026-07-24, the date on which the Version 1.0 reports were approved, in place of the date each plan was approved. The revision history of this plan now gives the approval date.

8. It states campaign thresholds for the significance of model terms (Section 6), for model adequacy (Section 7.2) and for capability (Section 7.9). Version 1.0 left all three to the individual study.

In Version 1.0 the cation exchange pool was judged against an aggregate limit of 1 % HMW. That limit was the drug substance criterion of 5 % divided by a margin of 5. The derivation in this plan credits no step after cation exchange with aggregate removal. The cation exchange pool therefore carries the drug substance criterion itself, 5 % (Table 7.1). The Version 1.0 limit was never an approved criterion, because it was applied at analysis and appeared in no approved plan, so the wider criterion now in force replaces no approved value. What moved at each step as a result is stated in the “Changes in this revision” section of its PCR-00N, from its own data.

2 Stage 1 characterization strategy

Process validation follows a lifecycle, and its first stage is process design (U.S. Food and Drug Administration 2011). In Stage 1 the commercial process is defined from the knowledge gained in development, and the studies that support it are documented. This campaign follows the enhanced approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012). Prior knowledge and risk assessment identify which parameters may affect which attributes. Characterization studies then measure those effects and bound them. The result is a set of normal operating ranges (NORs), PARs and parameter classes, from which the control strategy is built (CMC Biotech Working Group 2009).

A-Mab is a humanized IgG1 antibody developed on the Novacyte platform. The downstream train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, virus filtration and UF/DF was used for three earlier antibodies, X-Mab, Y-Mab and Z-Mab. Their history establishes the mechanism of each step and the range of performance to expect from it. Characterization of A-Mab therefore confirms and bounds known behaviour. The same history supplies the worst case of each downstream step, on which the intermediate acceptance criteria are built (Section 7.3).

Criticality is treated as a continuum, as described in ICH Q8(R2) (International Council for Harmonisation 2009). RA-001 ranks each parameter and assigns it to multivariate study, to univariate study or to no new study (Section 4). Parameters assigned to multivariate study are characterized by designed experiments in a qualified scale-down model. Such designs are planned for 6 steps, from the production bioreactor to virus filtration. The harvest (Step 4) and UF/DF (Step 10) steps are characterized by univariate and qualitative studies only, because RA-001 assigned none of their parameters to multivariate study.

Against the NOR in force (Table 4.2), the characterized range is wider than the NOR at both ends for 32 of the 37 parameters. For 4 parameters (post-clarification turbidity, filtration volume (load), number of diavolumes and transmembrane pressure) the low end of the characterized range equals the low end of the NOR, and only the high end extends beyond it. For A-Mab concentration the NOR equals the characterized range. Where the range is wider, it shows how the process behaves near and beyond the edges of routine operation. It also gives the knowledge space within which a PAR can be claimed. A PAR is claimed only where the data support it, and never beyond the range studied.

Viral safety rests on modular and orthogonal clearance across low-pH inactivation, anion exchange and virus filtration (International Council for Harmonisation 2023). Each of these steps is studied in its own spiking model, and each step plan claims only the log reduction of its own step. The cumulative claim is consolidated in PCMR-001.

The outputs of Stage 1 feed process performance qualification at the receiving site. PTP-001 plans 5 PPQ batches. The ranges and classes that this campaign supports are the ones those batches will be run against.

3 Critical quality attribute framework

The CQAs and their drug substance acceptance criteria are taken from the A-Mab case study (CMC Biotech Working Group 2009). Several of the criteria were set there from clinical experience with X-Mab, the closest platform antibody. They are held in the drug substance CQA register, and Table 3.1 lists all 11 attributes. Criticality was assigned with Tool #1 of the case study, as the product of impact and uncertainty. The levels run from very low (VL) to very high (VH), and the Tool #1 score stands beside each level. For every attribute except viral clearance, a higher score gives an equal or higher level. The two viral clearance attributes were assigned very high criticality by rule, because a viral contamination would directly harm the patient, whatever the uncertainty term of the score gives.

Table 3.1: Drug substance CQA register, with acceptance criteria, criticality and the step at which each attribute is formed or set.

Category	CQA	Acceptance	Criticality	Tool #1 score	Formed or set at
glycosylation	Afucosylation	2–13 % of glycans	H	60	Production Bioreactor
	Galactosylation (total %Gal)	10–40 % (G1+G2 equiv.)	H	48	Production Bioreactor
	High Mannose	3–10 % of glycans	VH	80	Production Bioreactor
size variant	Aggregates (HMW)	≤ 5 % HMW (SEC)	H	60	Production Bioreactor
	Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)	L	12	Harvest and Clarification
charge variant	Acidic charge variants (deamidation)	18–40 % (CEX/icIEF)	VL	4	Production Bioreactor
process impurity	Host Cell Protein (HCP)	≤ 100 ng/mg	M-H	36	Production Bioreactor
	Residual DNA	≤ 0.001 ng/dose	VL	6	Production Bioreactor
	Leached Protein A	≤ 5 ppm	L-M	16	Protein A Chromatography
viral safety	Viral clearance — XMuLV (enveloped)	≥ 16.7 log10 (cumulative)	VH	20	Low-pH Viral Inactivation
	Viral clearance — MVM (parvovirus)	≥ 8.6 log10 (cumulative)	VH	20	Anion Exchange Chromatography

The last column of Table 3.1 names the step at which each attribute is formed or set. Most attributes are formed in the production bioreactor. Afucosylation, galactosylation and high mannose are formed there, and no downstream step changes them. Protein A binds the Fc region at the interface of the CH2 and CH3 domains, so it captures every glycan variant alike. The two ion exchange steps do not separate glycan variants under platform conditions (CMC Biotech Working Group 2009). Leached Protein A enters the process at capture. Fragments are formed in culture and during the harvest hold, by proteases released from lysed cells that clip the antibody. The register therefore assigns them to harvest. No criterion for fragments is set at harvest, and they are judged at the downstream steps (Table 7.1). They are not an attribute of the production bioreactor designs (Table 7.3).

The two viral clearance attributes are cumulative. The register assigns each to one step, but each requirement is met by the sum of the claims of several steps (Section 7.4).

The attributes of very low criticality (acidic charge variants (deamidation) and residual DNA) remain in the register, and their criteria apply.

The output of a step that removes an impurity, or that adds to a variant, is judged against an intermediate acceptance criterion derived from the drug substance criterion (Section 7.3).

4 Risk-based prioritization

RA-001 assessed every process parameter of the train before characterization began. Each parameter was ranked on its potential impact on the CQAs and on process performance, and on its potential to interact with other parameters. Parameters ranked high, or with a likely interaction, were assigned to multivariate study. Parameters ranked lower, whose range could be supported one factor at a time, were assigned to univariate study. RA-001 gives the scores and the rationale for each assignment, and they are not repeated here.

Table 4.1 shows the result across the train. Of the 37 parameters, 22 are studied in multivariate designs and 15 one factor at a time. All 6 parameters of harvest (Step 4) and UF/DF (Step 10) are studied univariately.

Table 4.1: Characterization scope per step, from RA-001.

Step	Unit operation	Parameters in scope	In multivariate designs	One factor at a time	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest / Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration (formulation)	3	0	3	PCP-010 / PCR-010

Table 4.2 lists every parameter with its set-point, NOR and characterized range. The NOR shown is the NOR in force. At Version 2.0 PCR-006 narrowed the NOR of inactivation pH, hold time and temperature at low-pH viral inactivation and moved the set-point of inactivation pH and hold time, and the table footnotes the rows that changed. The class column gives the classification assigned in each PCR-00N under the parameter classification procedure SOP-4001, by the rules in Section 7.7.

Table 4.2: Process parameters by step: set-point, ranges and classification.

Step	Parameter	Unit	Set-point	NOR	Characterized range	Study	Class
Production bioreactor	Culture pH	pH	6.85	6.75–6.95	6.6–7.1	multivariate	WC-CPP
	Culture temperature	°C	35	34.5–35.5	34–36	multivariate	WC-CPP
	Dissolved CO2 (pCO2)	mmHg	100	70–130	40–160	multivariate	WC-CPP
	Osmolality	mOsm	400	375–425	360–440	multivariate	WC-CPP
	Culture duration	days	17	16–18	15–19	multivariate	WC-CPP
	Dissolved oxygen	% sat	50	40–60	30–70	univariate[a)]	KPP
	Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	univariate[a)]	KPP
	Basal medium concentration	X	1.2	1–1.4	0.8–1.6	univariate[a)]	GPP
	Nutrient feed-1 volume	% of working volume	12	10–14	9–15	univariate[a)]	KPP
Harvest	Centrifugation (rcf)	g	9000	8000–10000	6000–12000	univariate[a)]	KPP
	Depth filter load	L/m2	120	100–140	80–160	univariate[a)]	KPP
	Post-clarification turbidity	NTU	5	1–10	1–20	univariate[a)]	GPP
Protein A	Protein load	g/L resin	30	20–40	10–50	multivariate	WC-CPP
	Elution buffer pH	pH	3.55	3.4–3.7	3.2–3.9	multivariate	WC-CPP
	Load flow rate	cm/hr	200	150–250	100–300	multivariate	KPP
	End of pool collect	CV	2.6	2.3–2.9	2–3.2	multivariate	KPP
	Operating temperature	°C	22	18–26	15–30	univariate[a)]	GPP
	Bed height	cm	20	15–25	10–30	univariate[a)]	GPP
Low-pH viral inactivation	Inactivation pH	pH	3.4[c2)]	3.3–3.5[c1)]	3.2–4	multivariate	CPP
	Hold time	min	105[c4)]	90–120[c3)]	60–180	multivariate	WC-CPP
	Temperature	°C	21	20–23[c5)]	15–25	multivariate	WC-CPP
	A-Mab concentration	g/L	20	as characterized[b)]	5–35	univariate[a)]	GPP
Cation exchange	Protein load	g/L resin	25	18–32	10–40	multivariate	WC-CPP
	Load/Wash conductivity	mS/cm	5	4–6	3–7	multivariate	WC-CPP
	Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	multivariate	WC-CPP
	Elution stop collect	AU (OD280, descending)	1	0.7–1.3	0.5–1.5	multivariate	WC-CPP
	Elution flow rate	cm/hr	200	150–250	100–300	univariate[a)]	GPP
Anion exchange	Load pH	pH	7.5	7.35–7.65	7.2–7.8	multivariate	WC-CPP
	Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	multivariate	WC-CPP
	Load conductivity	mS/cm	5.5	4–7	3–8	multivariate	WC-CPP
	Protein load	g/L resin	200	120–280	50–300	multivariate	WC-CPP
	Operating flow rate	cm/hr	300	200–400	150–450	univariate[a)]	WC-CPP
Virus filtration	Filtration volume (load)	L/m2	90	50–105	50–140	multivariate	WC-CPP
	Filtration pressure	psi	13	10–20	8–30	multivariate	WC-CPP
UF/DF	Number of diavolumes	DV	8	7–9	7–10	univariate[a)]	KPP
	Transmembrane pressure	psi	12	10–14	10–15	univariate[a)]	KPP
	Final DS concentration	g/L	75	70–80	65–85	univariate[a)]	KPP

[a)] Studied one factor at a time.
[b)] The normal operating range equals the characterized range.
[c1)] Narrowed at Version 2.0 from pH 3.4–3.6, the NOR as characterized (PCR-006).
[c2)] Moved at Version 2.0 from pH 3.5, the set-point the studies were run around (PCR-006).
[c3)] Narrowed at Version 2.0 from 60–120 min, the NOR as characterized (PCR-006).
[c4)] Moved at Version 2.0 from 90 min, the set-point the studies were run around (PCR-006).
[c5)] Narrowed at Version 2.0 from 19–23 °C, the NOR as characterized (PCR-006).

The range of a parameter studied univariately is supported by its own data, and its class is decided by the same rules as for a parameter studied in a design.

5 Scale-down model strategy

Every characterization study is run in a scale-down model. The model is qualified under the scale-down model procedure SOP-1001 before its first characterization run. It is designed on the scaling principles of its unit operation. Parameters that do not depend on scale are held at their commercial values. Parameters that depend on scale are set so that the model matches the performance of the manufacturing scale.

For the chromatography steps the model keeps the bed height and the residence time of the manufacturing column. Load, wash and elution volumes are normalized to column volumes, and column efficiency is checked by plate count and peak asymmetry. For the production bioreactor the model is a stirred tank with equivalent mixing, aeration and process control. In each study, the parameters not under study are held at their commercial set-points. The designs vary culture pH, culture temperature, dissolved CO2 (pCO2), osmolality and culture duration, and univariate studies vary dissolved oxygen, initial viable cell conc., basal medium concentration and nutrient feed-1 volume. Agitation and gas flow are set to match the oxygen transfer and carbon dioxide stripping of the commercial vessel. For virus filtration the model uses the same filter type as manufacturing, and load is scaled by membrane area. For UF/DF, membrane loading and cross flow are scaled by membrane area.

Qualification compares the scale-down model, run at set-point, with manufacturing-scale batches. The outputs compared are those of the production bioreactor (afucosylation, galactosylation, high mannose, acidic variants, aggregate), Protein A (host cell protein, step yield, leached Protein A, plate count, peak asymmetry), low-pH viral inactivation (acidic variants, aggregate), cation exchange (step yield, plate count, peak asymmetry, aggregate, host cell protein), anion exchange (step yield, plate count, peak asymmetry, host cell protein) and virus filtration (step yield). A scale-down model need not reproduce the manufacturing mean. The difference between the two scales must instead be known precisely enough that no conclusion depends on its exact size. The rule is in Section 7.1, and each PCP-00N applies it to its own outputs. A model qualified for one output is not by that fact qualified for another. Where a model cannot be qualified for an output, the PCP-00N should say so and should state how that output is characterized instead.

Viral clearance is studied in a spiking model of each claimed step, because virus cannot be introduced at manufacturing scale. The spiking model follows the same scaling principles as the characterization model of the step (International Council for Harmonisation 2023).

The campaign used 3 instrumented scale-down systems under calibration and change control. They are the temperature-controlled chromatography chamber (EQ-CHR-118), the production scale-down bioreactor pCO2 probe (EQ-BRX-205) and the ultrafiltration / diafiltration TFF skid (EQ-TFF-142). Table 5.1 lists each with the studies

that used it. Every study window closed before the next calibration due date of the equipment it used. Every study that the Version 1.0 reports analysed closed before those reports were approved on 2026-07-24. The last of them ended on 2026-05-15. One supplementary qualification study of the Protein A scale-down model, S-PA-SDM2 (2026-09-07 to 2026-09-11), ran afterwards, while INV-001 was open. Its runs were made to narrow the confidence interval of the offset for host cell protein in the Protein A pool (Section 7.1). They are work that the Protein A plan PCP-005 did not foresee, and the Protein A report PCR-005 records them as discrepancy PD-005-01 from that plan.

Table 5.1: Execution windows of the studies that used instrumented systems, against the calibration due date and the date in the Version 1.0 title blocks.

Equipment	Description	Next calibration due	Study executed on the system			Date in Version 1.0 title block
			Study	Start	End	
EQ-CHR-118	temperature-controlled chromatography chamber	2026-05-30	S-CEX-RSM	2026-03-02	2026-03-27	2026-07-24
EQ-BRX-205	production scale-down bioreactor pCO2 probe	2026-06-14	S-BRX-SDM S-BRX-SCR S-BRX-UNI S-BRX-RSM	2025-08-27 2025-09-08 2025-10-06 2025-10-20	2025-09-05 2025-10-03 2025-10-17 2025-12-12	2026-07-24
EQ-TFF-142	ultrafiltration / diafiltration TFF skid	2026-04-22	S-UFDF-SDM S-UFDF-UNI	2026-02-25 2026-03-09	2026-03-06 2026-03-27	2026-07-24

Intermediate pools are held between steps. During a hold, aggregate can form, asparagine residues can deamidate and raise the acidic variants, and proteases in the clarified harvest can clip the antibody into fragments. Each of these raises its attribute with time. The campaign includes 3 hold studies. The clarified harvest and the Protein A eluate (at elution pH) are each held at 2-8 °C and 21 °C, and the neutralized viral inactivation pool at 21 °C and 25 °C, for up to 7 days. The design of the viral inactivation pool study follows A-Mab Table 4.17 (CMC Biotech Working Group 2009). Each hold study is planned in the PCP-00N of the step that produces the pool.

Each response is measured by a validated method under its own SOP. Table 5.2 lists every controlled procedure and method validation cited in the package. The SOP and AMV numbers are placeholders.

Table 5.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2003	Operation of the Production Bioreactor (Fed-Batch)
	SOP-2004	Cell-Culture Media and Feed Preparation
	SOP-2005	Harvest by Continuous Disk-Stack Centrifugation
	SOP-2006	Clarification by Depth and Sterile Filtration
	SOP-2007	Operation of the Protein A Capture Chromatography Step
	SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization
	SOP-2009	Operation of the Low-pH Viral Inactivation Step
	SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step
	SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step
	SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step
	SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step
	SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC
	SOP-3011	Size-Variant Analysis by SEC-HPLC
	SOP-3012	Host-Cell-Protein Quantitation by ELISA
	SOP-3013	Charge-Variant Analysis by icIEF
	SOP-3014	Residual Host-Cell DNA by qPCR
	SOP-3015	Purity by Non-Reduced CE-SDS
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3013	Charge Variants (icIEF)
	AMV-3014	Residual DNA (qPCR)
	AMV-3015	Turbidity by Nephelometry (NTU)
	AMV-3016	Leached Protein A by ELISA (ppm)
	AMV-3017	XMuLV Infectivity Titre (TCID50)
	AMV-3018	MVM Infectivity Titre (TCID50/qPCR)
	AMV-3019	Protein Concentration by A280 (UV)
	AMV-3020	Purity (non-reduced CE-SDS)

6 Common statistical approach

6.1 Designs

Each step with a multivariate scope is studied in two stages, under the design of experiments procedure SOP-1002. A screening design comes first. It is a two-level factorial design, full or fractional as the number of factors requires, with replicated centre points. A response-surface design follows. It is a central composite design over the factors that screening showed, or RA-001 expected, to affect a CQA with a criterion at the step. It also has replicated centre points. Table 6.1 gives the structure of each design. Every design has at least 3 centre points. The centre points estimate pure error and allow a test for curvature and for lack of fit.

Table 6.1: Designed experiments per step. Runs are given with the number of centre points in brackets.

Step	Unit operation	Screening			Response surface		Responses
		Factors	Runs	Model terms	Factors	Runs	
3	Production Bioreactor	5	19 (3)	16	4	28 (4)	Afucosylation, Galactosylation, High mannose, Acidic variants, Aggregates (HMW)
5	Protein A	4	19 (3)	11	4	28 (4)	Pool HCP (ng/mg), Step yield, Leached Protein A (ppm), Fragments (LMW, %, nrCE-SDS)
6	Low-pH Viral Inactivation	3	11 (3)	7	3	18 (4)	Aggregates (HMW, %), XMuLV LRF ($\log_{10}$), Acidic variants, Fragments (LMW, %, nrCE-SDS)
7	Cation Exchange	4	19 (3)	11	4	28 (4)	Aggregates (HMW, %), Pool HCP (ng/mg), Step yield, Fragments (LMW, %, nrCE-SDS)
8	Anion Exchange	4	19 (3)	11	4	28 (4)	Pool HCP (ng/mg), XMuLV LRF ($\log_{10}$), MVM LRF ($\log_{10}$), Step yield, Fragments (LMW, %, nrCE-SDS)
9	Small-Virus Retentive Filtration	2	7 (3)	4	2	12 (4)	MVM LRF ($\log_{10}$), XMuLV LRF ($\log_{10}$), Step yield, Fragments (LMW, %, nrCE-SDS)

Runs are executed in random order within each design. Where a run is lost or invalidated, the PCR-00N should report it and state whether the design was repaired or re-executed.

6.2 Models, and what each model is for

The screening data are fitted with a model of main effects and two-factor interactions in coded units. The response-surface data are fitted with a full quadratic model. Effects are reported as the change in the response over the coded range of the factor, and a model term is significant when its p-value is below $\alpha = 0.05$. This is the conventional level. Its consequence is that about one term in twenty with no real effect is called significant by chance. An attribute is called affected by a parameter when a term that contains the parameter is significant at that level.

The screening model identifies effects. Its centre points detect curvature but cannot attribute it to a factor, and the model still interpolates linearly between the corners of the design. In the production bioreactor and small-virus retentive filtration screening designs the interaction model also leaves only 1 degree of freedom for lack of fit (Table 6.1), so its fit statistics overstate how well it fits. The response-surface model is the predictive model. Every prediction, every PAR and every design space in the package comes from it. No report should claim prediction from a screening model.

A response-surface model predicts the mean response over the range studied. At the edge of an acceptable region defined on mean predictions, a single batch meets its criterion about half the time if variability is symmetrical. For this reason a quality attribute is judged on a single batch, not on the mean prediction, in two analyses. The NOR-propagated proven acceptable range (Section 6.4) is computed by Monte Carlo, and its interval includes the model residual, the uncertainty of the fitted coefficients and the variation of the other parameters within their NOR. The judgement of an output over the NOR in the scale-down qualification (Section 7.1) uses the exact 95 % prediction interval of a single run at each combination of NOR low, set-point and NOR high.

6.3 Model adequacy

Each fitted model is reported with R^2, adjusted R^2 and predicted R^2, the model F-test, and the lack-of-fit test against pure error from the centre points. Residuals are inspected against predicted values and on a normal probability plot. A model is used for prediction only if it meets the adequacy criteria in Section 7.2.

6.4 Proven acceptable ranges

A PAR is computed for each CQA that has a criterion at the step and for each factor of the response-surface model.

1. The set-point analysis is a univariate proven acceptable range with the other parameters at their set-points. It scans the factor across its characterized range. The PAR is the contiguous range around the set-point over which the mean prediction meets the criterion.

2. The NOR-propagated analysis is the same range with the other parameters varying within their NORs. It scans the factor across 81 grid points. The PAR is the contiguous range around the set-point over which the predicted result meets the criterion.

What the NOR-propagated analysis compares with the criterion depends on the kind of attribute.

- For a quality attribute it compares the 95 % prediction interval of a single batch, because a single batch is what must meet the criterion. At each grid point the analysis draws 2,000 operating points for the other factors. Each factor is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, and clipped to the characterized range. This assumes that the NOR spans ±3 standard deviations of the routine variation of the parameter. The interval includes the model residual and the uncertainty of the fitted coefficients. It runs from the 2.5th to the 97.5th percentile of the simulated batches. For an upper limit the 97.5th percentile must not exceed the limit, for a lower limit the 2.5th percentile must not fall below it, and for a two-sided criterion both apply.
- For viral clearance it compares the worst case within the NOR. At each grid point the other factors are set to every combination of their NOR low, set-point and NOR high. The lowest mean predicted log reduction among these combinations must meet the claim of the step. Viral clearance is validated at worst-case conditions, so a claim that held only on average over the NOR would not support a validated log reduction.

The NOR-propagated PAR is the one that supports a claim. It is in general narrower than the set-point PAR, because the range must hold while the rest of the operating point moves within its NOR.

6.5 Capability, and the integrated model

Capability at commercial scale is estimated from an integrated model of the train. The model runs 2,000 simulated batches through every step in sequence. Each process parameter of each batch is drawn from the same normal distribution as in the NOR-propagated analysis of a quality attribute (Section 6.4). The drug substance attributes of each batch are recorded.

Capability is reported as Cpk against the drug substance criterion. For a criterion with one limit, Cpk is computed against that limit. For a two-sided criterion it is the lower of the two one-sided indices. Acidic variants have a two-sided criterion (18 to 40 %), and their capability is assessed against both ends. The criteria for capability are in Section 7.9.

The same simulation gives the probability of an out-of-specification drug substance for each attribute. Two figures are reported, because a few thousand simulated batches cannot resolve a rare event. These are the count of simulated batches outside the criterion, and the tail probability of a normal distribution with the simulated mean and

standard deviation, one- or two-sided as the criterion is. The intermediate acceptance criteria judge each step separately, and the integrated model shows whether the steps together deliver the drug substance.

7 Acceptance criteria framework

Each PCP-00N applies the criteria below to its own step. They define when a scale-down model is qualified, a model adequate, a step output acceptable, a range proven and a parameter classified. Where a value depends on the step, this plan defines the form of the criterion and the step plan sets its value. In Version 1.0 each step plan was approved before its studies ran. The a priori derivation of the Version 2.0 criteria came after the studies, and for that reason it uses platform data and the drug substance criteria only (Section 7.3). Verification then replaced some platform values with A-Mab confidence bounds, and 5 of the 22 criteria in force include such a bound (Section 7.5).

7.1 Scale-down model qualification

A scale-down model is qualified for each output separately. It is qualified by the offset between the scales, not by a test of equivalence. A model may differ in mean from manufacturing scale, provided the difference is known precisely enough. The rule has four parts.

1. The offset of an output is the mean of the scale-down runs at set-point minus the mean of the manufacturing-scale batches. It is reported with its two-sided 90 % confidence interval. Each end of that interval is a one-sided 95 % confidence bound, which is the level used to compare two means at 5 % on each side.
2. Where the qualification compared an output at both scales, every judgement of a scale-down result for that output against a criterion first subtracts its offset. Ranges and conclusions are therefore stated at manufacturing scale.
3. The model is qualified for an output when the conclusion against its criterion over the NOR is the same with the offset at either end of its confidence interval. The conclusion is whether the output meets its criterion at every combination of NOR low, set-point and NOR high of the response-surface factors.
4. An output with no criterion at the step, such as step yield, has its offset recorded only. No conclusion depends on it.

Where the cause of an offset is known, the report states it. A model whose conclusion for an output changes between the two ends of the offset interval is not used for that output. The qualification studies do not compare fragments or viral clearance at both scales. Their offset is unknown, not zero. These outputs are judged as the model predicts them because the qualification did not compare them, and their representativeness rests on what each step plan states, which is the mechanism of the step for fragments and the spiking model for viral clearance.

Each PCP-00N names the outputs, the number of runs at each scale and the manufacturing-scale data set before its first characterization run. Each PCR-00N tabulates the offsets of its step, and PCMR-001 summarises them for the whole train.

7.2 Model adequacy

A response-surface model is adequate for prediction when the following hold.

1. The model F-test is significant at the level set for model terms, $\alpha = 0.05$ (Section 6).
2. Lack of fit against pure error is not significant at $\alpha = 0.05$.
3. Predicted R^2 falls short of R^2 by no more than 0.20. This is a limit on overfitting, because a model that fits its own runs much better than it predicts left-out runs has more terms than the data support.
4. The residuals show no pattern against predicted values or run order, and no single run dominates the fit.

A model that fails the lack-of-fit test or the limit on overfitting is either refitted with a justified change of terms or transformation, or it is not used to set a range.

Where no adequate model results, the PCR-00N should report the response descriptively, against the criterion, over the runs of the design.

7.3 Intermediate acceptance criteria

A drug substance criterion does not say what the output of an upstream step must meet. Protein A, for example, leaves host cell protein in its pool that cation exchange and anion exchange will remove. The Protein A pool is expected to carry far more host cell protein than the drug substance limit of 100 ng/mg, so that limit cannot judge it. Each step output is therefore judged against an intermediate acceptance criterion.

The rule is the following. Each drug substance criterion is carried backward through the train, step by step, using the worst case that each downstream step showed on the platform. For a step that removes an attribute, the worst case is the lowest clearance fold observed for X-Mab, Y-Mab and Z-Mab. For a step that adds to an attribute, it is the largest increase observed. No margin is applied. An attribute that no downstream step changes keeps its drug substance criterion. This is the case for afucosylation, galactosylation and high mannose.

For host cell protein the rule gives the Protein A pool an a priori criterion of 16,000 ng/mg. The drug substance limit of 100 ng/mg is carried back through the platform worst case of anion exchange (4-fold) and then of cation exchange (40-fold). The cation exchange pool is judged against the limit carried back through anion exchange alone, 400 ng/mg a priori. For A-Mab, the 95 % confidence bounds of the host cell protein clearance were 41.8-fold for cation exchange and 5.18-fold for anion exchange.

Both are better than the platform worst cases. Both host cell protein models meet the model adequacy criteria, so both bounds replaced the platform values (Section 7.2, Section 7.5). The Protein A pool criterion is therefore 21,662 ng/mg, and the cation exchange pool criterion is widened to 518 ng/mg.

Table 7.1 gives all 22 criteria across 6 steps, with the downstream worst case each one is derived from. The downstream worst case shown is the cumulative fold or increase of all steps after the step output. In a tightened or a widened row it includes the A-Mab confidence bound that replaced a platform worst case (Section 7.5). The footnotes of the table mark these rows. The source of each platform value is recorded in the configuration of the process model and in the verification table of each report.

Table 7.1: Intermediate acceptance criteria and their derivation from the drug-substance criteria.

Attribute	DS criterion	Step output	Downstream worst case		IAC	Basis
			Clearance fold	Increase		
Host cell protein (ng/mg)	≤ 100	Protein A	217	0.00	≤ 21,662	Carried back, widened[a]
		Cation exchange	5.18	0.00	≤ 518	Carried back, widened[a]
		Anion exchange	1.00	0.00	≤ 100	DS criterion[b]
Aggregates, HMW (%)	≤ 5.00	Production bioreactor	1.52	0.53	≤ 7.04	Carried back, tightened[c] and widened[a]
		Low-pH viral inactivation	1.52	0.00	≤ 7.58	Carried back, tightened[c]
		Cation exchange	1.00	0.00	≤ 5.00	DS criterion[b]
Fragments, LMW (%)	≤ 2.00	Protein A	1.30	0.15	≤ 2.45	Carried back
		Low-pH viral inactivation	1.30	0.00	≤ 2.60	Carried back
		Cation exchange	1.00	0.00	≤ 2.00	DS criterion[b]
		Anion exchange	1.00	0.00	≤ 2.00	DS criterion[b]
		Virus filtration	1.00	0.00	≤ 2.00	DS criterion[b]
Leached Protein A (ppm)	≤ 5.00	Protein A	31.6	0.00	≤ 158	Carried back
Acidic variants (%)	18.0–40.0	Production bioreactor	1.00	0.40	18.0–39.6	Carried back, widened[a]
		Low-pH viral inactivation	1.00	0.00	18.0–40.0	DS criterion[b]
Afucosylation (%)	2.00–13.0	Production bioreactor	1.00	0.00	2.00–13.0	DS criterion[b]
Galactosylation (%)	10.0–40.0	Production bioreactor	1.00	0.00	10.0–40.0	DS criterion[b]
High mannose (%)	3.00–10.0	Production bioreactor	1.00	0.00	3.00–10.0	DS criterion[b]
XMuLV clearance (log10)	≥ 16.7[d]	Low-pH viral inactivation	not applicable		≥ 6.60	Platform claim[e]
		Anion exchange	not applicable		≥ 5.50	Platform claim[e]
		Virus filtration	not applicable		≥ 4.62	Platform claim[e]
MVM clearance (log10)	≥ 8.60[d]	Anion exchange	not applicable		≥ 4.00	Platform claim[e]
		Virus filtration	not applicable		≥ 4.62	Platform claim[e]

[a] The 95 % confidence bound of a downstream step's A-Mab worst case within its NOR was better than the platform worst case. The bound replaced it.
[b] No downstream step changes the attribute, so the drug-substance criterion applies.
[c] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The 95 % confidence bound of that prediction replaced it.
[d] Cumulative requirement across the steps that clear the virus.
[e] Log reduction claimed for this step in the modular viral clearance claim.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

The rule above credits each downstream step only with the clearance or increase in its platform history and assumes no other change in its operation. Note b) of Table 7.1 marks the rows where no downstream step is credited, so the drug substance criterion applies. A hold between steps can still raise aggregate and acidic variants (Section 5). Holds are covered by the hold studies, each planned in the PCP-00N of the step that produces the pool and judged against a hold criterion set in that plan.

Residual DNA has a drug substance criterion but no intermediate acceptance criterion. The Protein A designs measure DNA in the load and the pool, but the platform history used here gives no worst-case DNA clearance for the steps after Protein A, so no criterion can be carried back. Residual DNA is therefore judged on the drug substance only, and its capability is assessed in the integrated model (Section 6.5).

Of the 22 criteria in Table 7.1, 17 are in force at their a priori value and verification changed 5. The tightened criteria are the aggregate criteria upstream of cation exchange, because the cation exchange model gave a lower aggregate clearance than the platform worst case. One of them, the criterion for aggregate in the production bioreactor harvest (from 9.30 % to 7.04 %), was also widened, because the low-pH viral inactivation model gave a smaller aggregate increase (0.54 % at the 95 % bound, against 0.70 % on the platform). The result is a stricter criterion than the a priori one. The other tightened criterion is the criterion for aggregate in the viral inactivation pool (from 10.0 % to 7.58 %). The criteria only widened are the upper limit for acidic variants in the production bioreactor harvest (from 39.0 % to 39.6 %), the criterion for host cell protein in the Protein A pool (from 16,000 ng/mg to 21,662 ng/mg) and the criterion for host cell protein in the cation exchange pool (from 400 ng/mg to 518 ng/mg). The report that relies on a widened criterion gives the argument for it (Section 7.5).

Applying these criteria to studies already executed is acceptable because the a priori derivation uses no data from the studies it judges. The drug substance criteria come from the CQA register. The worst cases come from the platform history of X-Mab, Y-Mab and Z-Mab. A-Mab data enter a criterion only when verification replaces a platform worst case with the 95 % confidence bound of the A-Mab worst case of the same step (Section 7.5).

7.4 Viral clearance claims

Each step that clears virus is held to its platform claim. The claim is the log reduction that the step has delivered for that virus on the platform, with X-Mab, Y-Mab and Z-Mab. The sum of the claims is then checked against the cumulative requirement of the drug substance. Table 7.2 shows that check for each model virus. Each step is then judged against its own claim at the worst case within its NOR, as Section 6.4 defines it.

Table 7.2: Platform claims per step (log10) and their sum against the cumulative requirement of the drug substance.

Virus	Low-pH viral inactivation	Anion exchange	Virus filtration	Sum of claims	Cumulative requirement	Met
XMuLV (enveloped)	6.6	5.5	4.62	16.72	16.7	yes
MVM (parvovirus)	no claim	4	4.62	8.62	8.6	yes

The claims for XMuLV sum to 16.72 log10 against a requirement of 16.7. For MVM they sum to 8.62 against 8.6. Both requirements are met by the claims alone, with little room above them. A step that falls short of its claim therefore cannot be made up by the surplus of another step. Low-pH viral inactivation makes no claim for MVM, because a parvovirus has no lipid envelope for low pH to disrupt.

7.5 Verification of the platform worst cases

Each intermediate acceptance criterion assumes that every downstream step performs at least as well as its platform worst case. That is, the step clears at least the lowest fold, or adds no more than the largest increase, seen for X-Mab, Y-Mab and Z-Mab. Each report checks the platform worst cases of its own step. The check uses the response-surface model of the step. The model gives the worst clearance fold, or the largest increase, that it predicts within the NOR, and the one-sided 95 % confidence bound of that worst case, on its unfavourable side. An assumption is confirmed when the predicted worst case is no worse than the platform worst case. The values observed at the edges of the characterized range are reported beside it for information.

Verification may change a criterion in either direction, by the following rule.

1. Where the A-Mab worst case predicted within the NOR is worse than the platform worst case, its 95 % confidence bound replaces the platform value. Every criterion upstream of that step is recalculated with it and becomes stricter.
2. Where the 95 % confidence bound of the A-Mab worst case within the NOR is better than the platform value, and the response-surface model behind it meets the adequacy criteria of Section 7.2, the bound replaces the platform value. Every criterion upstream of that step is recalculated with it and becomes wider.
3. Otherwise the platform value stands.

In both directions the bound, on its unfavourable side, replaces the platform value, and the point prediction never does.

A report that relies on a widened criterion gives the argument for it, with the bound, the runs behind it, and the reason why the performance of the downstream step for A-Mab is known well enough to replace the platform value. The rule was written into this plan before any Version 2.0 step report used it. This reissue of the plan also records what the rule gave, in Section 7.3 and below.

The criteria assume 9 platform worst cases. Of these, 7 are responses of the response-surface designs of low-pH viral inactivation, cation exchange and anion exchange, and they can be checked. The other 2 are the leached Protein A clearance of cation exchange and of anion exchange. Leached Protein A is not a response of those designs, so the platform values stand. The 95 % confidence bound was better than the platform worst case for the aggregate increase of low-pH viral inactivation, the acidic variants increase of low-pH viral inactivation, the host cell protein clearance of cation exchange and the host cell protein clearance of anion exchange, and these bounds replaced the platform values. For the fragments increase of low-pH viral inactivation and the fragments clearance of cation exchange the bound was also better, but the model does not meet the adequacy criteria, so the platform values stand. Verification found that the aggregate clearance of cation exchange within its NOR is lower than the platform worst case. That tightened the aggregate criteria upstream of cation exchange (Section 7.3). The cation exchange report PCR-007 gives the values and their analysis.

7.6 Proven acceptable ranges

A range of a parameter is proven acceptable for an attribute when the NOR-propagated analysis (Section 6.4) meets the criterion of that attribute across it. The criterion is the IAC where Table 7.1 defines one, and the drug substance criterion otherwise. A parameter that affects several attributes has a PAR for each, and the range carried into the control strategy is their intersection. Where the output does not meet its criterion even at the set-point, the PCR-00N reports that no PAR exists for the attribute and states the consequence for the NOR.

A PAR that reaches the edge of the characterized range is reported as reaching it, and no claim is made beyond it.

7.7 Parameter classification

Parameters are classified under the parameter classification procedure SOP-4001 by the decision logic of the A-Mab case study (CMC Biotech Working Group 2009).

1. A critical process parameter (CPP) has an effect on a CQA that is significant and practically relevant within the characterized range, and a high risk of falling outside its acceptable range in operation.
2. A well-controlled critical process parameter (WC-CPP) has the same kind of effect on a CQA but a low risk of falling outside its acceptable range, because it is readily controlled and its PAR is wide against its NOR.
3. A key process parameter (KPP) affects process performance, such as yield or concentration, and has no effect on a CQA within its range.
4. A general process parameter (GPP) has no relevant effect on a CQA or on process performance within its range.

A significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical. The PCR-00N should state the size of the effect beside its significance and give the reason for each class. The classification at this revision counts 1 CPP, 20 WC-CPP, 10 KPP and 6 GPP, and is consolidated in PCMR-001.

7.8 Method for fragments

From 2026-03-20, fragments are measured by nrCE-SDS (AMV-3020, under SOP-3015) in every study at every step. This follows the decision to change the fragment method to nrCE-SDS (AMD-001). One method across the train is needed because fragment results are compared across steps, and a comparison between two methods would mix method bias with step performance. Designs executed before that date measured fragments by SEC (Table 7.3). The screening designs of Protein A, low-pH viral inactivation and cation exchange and the invalidated first execution of the anion exchange design have SEC results only. The response-surface designs of Protein A and low-pH viral inactivation were measured by SEC, and their retained samples were later re-tested by nrCE-SDS. The cation exchange response-surface design ran across the decision date and measured fragments by both methods on every run.

The decision followed the comparability study S-CEX-FRC (2026-03-02 to 2026-03-13), in which SEC (AMV-3011) and nrCE-SDS were run on the same samples. SEC read fragments high wherever the sample still carried host cell protein, because the part of the host cell protein that elutes in the low molecular weight window of SEC is counted as fragment. The excess can never exceed the host cell protein present. It was a few tenths of a percent in the Protein A and viral inactivation pools and close to zero from the cation exchange pool onward. From the cation exchange pool onward the two methods agreed (INV-001 finding 6). SEC remains the method for aggregate (HMW), where no such interference applies.

Table 7.3 shows which method measured fragments in which study. Retained response-surface samples of Protein A and low-pH viral inactivation were re-tested by nrCE-SDS between 2026-04-06 and 2026-04-17. Screening samples were not retained. Their SEC fragment results are kept in the record but are not used for any decision. The re-executed anion exchange design ran after the decision and measured fragments by nrCE-SDS.

Table 7.3: Methods used for size variants in the designed studies, by step.

Step	Attribute	Method used in study	
		Screening	Response surface
Production bioreactor	Aggregates, HMW	SEC	
	Fragments, LMW	not in scope[a)]	
Harvest	Aggregates, HMW	no DoE[b)]	
	Fragments, LMW	no DoE[b)]	
Protein A	Aggregates, HMW	SEC	
	Fragments, LMW	SEC[c)]	SEC[c)], nrCE-SDS[d)]
Low-pH viral inactivation	Aggregates, HMW	SEC	
	Fragments, LMW	SEC[c)]	SEC[c)], nrCE-SDS[d)]
Cation exchange	Aggregates, HMW	SEC	
	Fragments, LMW	SEC	SEC, nrCE-SDS[e)]
Anion exchange	Aggregates, HMW	not measured[f)]	
	Fragments, LMW	nrCE-SDS[g)]	nrCE-SDS[g)]
Virus filtration	Aggregates, HMW	not measured[f)]	
	Fragments, LMW	nrCE-SDS	nrCE-SDS
UF/DF	Aggregates, HMW	no DoE[b)]	
	Fragments, LMW	no DoE[b)]	

[a)] Fragments were not an attribute of this study.
[b)] Univariate characterization only.
[c)] In this matrix part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[d)] Retained response-surface samples, tested 2026-04-06 to 2026-04-17.
[e)] Both methods on the same samples of this design.
[f)] Aggregate was not measured in the designs at this step.
[g)] Re-executed design (DEV-008-01). The invalidated first execution measured fragments by SEC.

7.9 Capability and the probability of an out-of-specification drug substance

Capability is estimated as in Section 6.5 and judged for each CQA in PCMR-001 against two values.

1. A Cpk of at least 1.33 is expected of a CQA in routine manufacture. At that value the nearest limit lies 3.99 standard deviations from the process mean, and a normal process would put about 33 batches per million beyond that limit.
2. A CQA with a Cpk below 1 needs an action in the control strategy. At that value the process spread, at three standard deviations on the limiting side, just reaches the criterion. The action may be a tighter NOR, an added in-process control or added testing. PCMR-001 should name the action.

A Cpk between the two values is acceptable. PCMR-001 should then state the side of the criterion that limits it and the parameters that drive it. For the probability of an out-of-specification drug substance, PCMR-001 should report the count of simulated batches outside each criterion and the tail probability, and should discuss every attribute with a simulated batch outside its criterion.

8 Register of characterization plans

Table 8.1 lists the step plans that apply this master plan, with the study type each one carries and its paired report.

Table 8.1: Process Characterization Plans under PCMP-001.

Plan	Subject	Study type	Paired report
PCP-003	Production Bioreactor (Step 3)	screening and response surface	PCR-003
PCP-004	Harvest and Clarification (Step 4)	univariate and qualitative	PCR-004
PCP-005	Protein A Chromatography (Step 5)	screening and response surface	PCR-005
PCP-006	Low-pH Viral Inactivation (Step 6)	screening and response surface	PCR-006
PCP-007	Cation Exchange Chromatography (Step 7)	screening and response surface	PCR-007
PCP-008	Anion Exchange Chromatography (Step 8)	screening and response surface	PCR-008
PCP-009	Small-Virus Retentive Filtration (Step 9)	screening and response surface	PCR-009
PCP-010	Ultrafiltration / Diafiltration (Step 10)	univariate and qualitative	PCR-010

Each step plan has the same sections, for the purpose and scope, the objectives, the prior knowledge and risk basis, the materials and methods, the study design, the acceptance and decision criteria, data management, roles, deliverables, and the risks and assumptions of the study. The criteria section of each plan states the criteria of Section 7 for its own outputs and adds the values that this plan leaves to the step.

9 Deliverables and schedule

The campaign delivers the following documents.

1. One PCP-00N and one PCR-00N for each of the 8 steps, each report paired with its plan.
2. PCMR-001, which consolidates the reports into the design spaces, the PARs, the parameter classes, the capability of the drug substance and the control strategy.
3. A ground-truth annex for each controlled document, built from the same data as the document.

In Version 1.0 each step plan was approved before the first characterization run of its step, and each report after the data of all its studies were complete. PCMR-001 is written last.

Table 9.1 records the execution window of every study, in order of start. The campaign ran from 2025-08-27 to 2026-09-11 across 34 studies. Some steps were studied in parallel, because their scale-down models are independent of each other. The first execution of the anion exchange design (S-AEX-SUP) was invalidated under deviation DEV-008-01, because its load material had been deamidated during an extended hold after neutralization and did not represent the process. The design was re-executed as S-AEX-SCR and S-AEX-RSM.

Table 9.1: Execution window of every characterization study.

Study	Step	Study type	Start	End
S-BRX-SDM	Production Bioreactor	scale-down model qualification	2025-08-27	2025-09-05
S-BRX-SCR	Production Bioreactor	screening design	2025-09-08	2025-10-03
S-BRX-UNI	Production Bioreactor	univariate	2025-10-06	2025-10-17
S-BRX-RSM	Production Bioreactor	response-surface design	2025-10-20	2025-12-12
S-HAR-UNI	Harvest and Clarification	univariate	2025-11-03	2025-11-21
S-PA-SDM	Protein A Chromatography	scale-down model qualification	2025-11-05	2025-11-14
S-PA-SCR	Protein A Chromatography	screening design	2025-11-17	2025-12-05
S-HAR-HLD	Harvest and Clarification	hold study	2025-11-24	2025-12-05
S-PA-UNI	Protein A Chromatography	univariate	2025-12-08	2025-12-12
S-VI-SDM	Low-pH Viral Inactivation	scale-down model qualification	2026-01-07	2026-01-16
S-PA-RSM	Protein A Chromatography	response-surface design	2026-01-12	2026-02-06
S-VI-SCR	Low-pH Viral Inactivation	screening design	2026-01-19	2026-01-30
S-PA-HLD	Protein A Chromatography	hold study	2026-02-02	2026-02-16
S-VI-RSM	Low-pH Viral Inactivation	response-surface design	2026-02-02	2026-02-13
S-VI-UNI	Low-pH Viral Inactivation	univariate	2026-02-02	2026-02-06
S-AEX-SDM	Anion Exchange Chromatography	scale-down model qualification	2026-02-04	2026-02-13
S-CEX-SDM	Cation Exchange Chromatography	scale-down model qualification	2026-02-04	2026-02-13
S-AEX-SUP	Anion Exchange Chromatography	first execution (invalidated)	2026-02-16	2026-03-04
S-CEX-SCR	Cation Exchange Chromatography	screening design	2026-02-16	2026-02-27
S-VI-HLD	Low-pH Viral Inactivation	hold study	2026-02-16	2026-02-27
S-UFDF-SDM	Ultrafiltration / Diafiltration	scale-down model qualification	2026-02-25	2026-03-06
S-CEX-FRC	Cation Exchange Chromatography	method comparability	2026-03-02	2026-03-13
S-CEX-RSM	Cation Exchange Chromatography	response-surface design	2026-03-02	2026-03-27
S-UFDF-UNI	Ultrafiltration / Diafiltration	univariate	2026-03-09	2026-03-27
S-CEX-UNI	Cation Exchange Chromatography	univariate	2026-03-16	2026-03-20
S-AEX-SCR	Anion Exchange Chromatography	screening design	2026-03-23	2026-03-31
S-AEX-RSM	Anion Exchange Chromatography	response-surface design	2026-04-01	2026-04-10
S-PA-RET	Protein A Chromatography	retest of retained samples	2026-04-06	2026-04-17
S-VI-RET	Low-pH Viral Inactivation	retest of retained samples	2026-04-06	2026-04-17
S-AEX-UNI	Anion Exchange Chromatography	univariate	2026-04-13	2026-04-17
S-VF-SDM	Small-Virus Retentive Filtration	scale-down model qualification	2026-04-15	2026-04-24
S-VF-DOE	Small-Virus Retentive Filtration	screening design	2026-04-27	2026-05-05
S-VF-RSM	Small-Virus Retentive Filtration	response-surface design	2026-05-06	2026-05-15
S-PA-SDM2	Protein A Chromatography	scale-down model qualification	2026-09-07	2026-09-11

10 Approvals

This plan is approved by the functions listed in the approval record at the front of this document, and each step plan is approved by the same functions.

Table 10.1: Responsibilities of the approving functions.

Function	Responsibility for this plan
Process Development / MSAT	Author. Owns the strategy, the designs and the acceptance framework
Analytical Development	Methods, method validations and the fragment method decision
Manufacturing Science	Scale-down model qualification and transfer to the receiving site
Statistics	Designs, models, PAR analysis and the integrated model
Quality Assurance	Approval, and closure of the actions from INV-001

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